Andrea Muolo

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EXPERIENCE

Senior .NET Developer and Business Analyst
Systemorph

Dec 2022 - present | Zürich, Switzerland

Developer of the Systemorph Cloud platform and client-tailored web-apps.

Experience with modern cloud architecture principles. Tech stack: .NET Core, .NET Compiler, C#, Entity Framework, MS SQL, SignalR, Kubernetes, Docker.

Led POCs and responsible for meeting clients, discussing business specifications, from data modelling, to business logic (e.g. actuarial methodologies, accounting standards), business processes, UX design.

Organize and conduct presentations to clients, on-site trainings, and workshops.

Developer and tester of in-house C# packages and concept applications to push the limits of our technology and foster our community of devs. Tech advocate.

.NET Developer and Business Analyst
Systemorph

Dec 2021 - Nov 2022 | Zürich, Switzerland

Receive business specifications from stakeholders and implement tailored data models, business formulas and processes in modular, scalable, and resilient web applications. Experience with actuarial methodologies and accounting standards.

Postdoctoral Research Fellow
Hebrew University of Jerusalem

Oct 2019 - Nov 2021 | Jerusalem, Israel

Exact Factorization techniques in molecular physics. Algebraic manipulation of exact factorization equation and numerical calculations to tackle non-adiabatic systems. Genuine predictions on the Berry phase. Methods developed in C++20.

Research and Teaching Assistant
ETH Zürich

Dec 2018 - Sept 2019 | Zürich, Switzerland

Many-body Dirac Equation and relativistic quantum mechanics. Novel analytical and numerical solutions to integrals to enable symmetry projected optimization.

Method development: Pre-Born–Oppenheimer and Tensor Networks algorithm.

Rich experience in concurrent and distributed computing using C++17.

EDUCATION

Doctor of Science Degree
ETH Zürich

Nov 2014 – Dec 2018 | Zürich, Switzerland

Explicitly Correlated Gaussians and the Quantum Few-Body Problem.

BSc and MSc in Physical Chemistry
University of Pavia and University of Cambridge

Sept 2009 – Sept 2014 | Pavia, Italy – Cambridge UK

MSc : A classical dynamic that conserves the quantum Boltzmann distribution.

BSc: Molecular Memories: rotaxane- and catenane-based molecular devices.

Second level MSc at Scuola Universitaria Superiore (IUSS)

MOST PROUD OF

Swiss National Science Foundation (SNSF) Fellowship
Research Grant, October 2019 - Nov 2021

Project Manager - from Jul 2023
Define scope, plan and organize the execution of the POC

TECH SKILLS

C++20 with libraries Armadillo, Boost, Eigen, and API OpenMP 4.0, MPI 3.1, SIMD, HPC ●●●●●

C#, .NET Core, Nuget, RestAPI, SignalR, Entity Framework, DI, ML.NET WPF, Razor, Blazor, MVC, MVVM, Kubernetes, Docker, CI/CD ●●●●●

Eclipse, Visual Studio, ReSharper, Git, Mathematica, LaTex ●●●●●

Javascript, Python, CSS, AJAX, JSON CSS, HTML5, VBA, OpenACC, CUDA ●●●●○

STRENGTHS

Algorithm Optimization | Strong Math Foundation | Concurrent Computation | Distributed Computing | Quantum Computing | Software Design | Scrum and agile framework | Cloud Architectures

Data Modelling | Digital Business Process | UX | Actuarial Methods | Accounting Standards IFRS17

Fast Learning | Scientific Writing | Teaching | Public Speaking | Cross Domain Knowledge

LANGUAGES

German	A2
English	Proficient
Italian	Mother Tongue

ABOUT ME

PUBLICATIONS SELECTION OF

J. Chem. Theory Comput., 18, 234 (2022)
Quantum Proton Effects from Density Matrix Renormalization Group Calculations

Phys. Rev. A, 102, 022803 (2020)
Analytically projected, rotationally symmetric, explicitly correlated Gaussians with one-axis shifted centers

J. Chem. Phys., 142, 134103 (2015)
Boltzmann-conserving classical dynamics in quantum time-correlation functions: “Matsubara dynamics”